Project 1: Website Development Agent Proposal

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Chosen Domain: Website Developement Agent

Goal of Agent: To generate a complete, structured HTML/CSS websites from natural language prompts.

Structured-Agent Architechture

1. Planner (LLM call):

Analyze your request

Identify website type and required section and produce structured layout

2. Generator (LLM call):

Generates semantic HTML

Generates semantic CSS styling

3. Validator (Rules and Optional LLM):

Check valid HTML scripts and remove unsafe scripts

Ensure required tags exist

4. Executor (Tool/API call):

Save HTML/CSS files

Package as downloadable as ZIP

Tools, Safety, Examples and Risks

1. Planned Tools and API:

OpenAI API (Planner and Generator)

Python file handling system

HTML/CSS validation libraries

2. Validation and safety:

Remove unsafe <script> tags

Enforce correct HTML structure

Limit file size

3. Example of Prompt:

Generate a restaurant homepage with menu section

4. Risks and Challenges

Broken HTML/CSS generation

Complex or ambiguous user requests

Responsive design inconsistencies

Thank You!